Scaling up RHD screening and prophylaxis repays its cost by 2050 in Indonesia, but not in Uganda

A modelled investment case for echo screening and secondary antibiotic prophylaxis

William Garcia

University of Washington · World Bank

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We compare scaled-up screening and prophylaxis with current care

Question

What do echo screening and benzathine penicillin G (BPG) prophylaxis buy by 2050, at what cost?

Evidence base

GBD 2023 burden; UN WPP 2024 population; WHF stages A–D (Rwebembera et al., 2023); investment-case design of Coates et al. (2021)

Scope

Deterministic scenario estimates for Indonesia and Uganda, 2026–2050.

About 2.2 million people in Indonesia and Uganda live with RHD, most of them undiagnosed

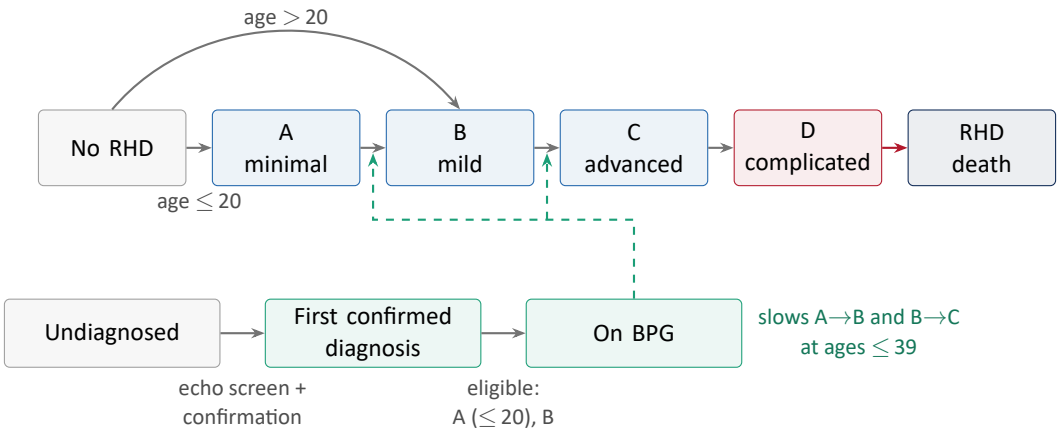
2.16 million

people living with rheumatic heart disease, 2023

Modelled share undiagnosed in 2026: 76% in Indonesia, 83% in Uganda

GBD 2023 published counts: 1,401,000 (Indonesia) and 763,000 (Uganda). Undiagnosed share: model reference arm, classifiable RHD at the end of 2026.

In the model, RHD kills only after progressing to stage D



Stages follow the WHF 2023 classification. Clinical detection also leads to diagnosis. Surgery is a service, equal in both arms. Other-cause death applies to every state.

Scale-up raises one-time screening to 80% and BPG initiation to 90% by 2050

Care input	Reference (held)	Scale-up, 2050
One echo screen across ages 5–39	5% / 1%	**80%**
BPG initiation among the diagnosed, per year	50%	**90%**
BPG retention, per year	90%	**95%**
Good adherence, share on BPG	60%	**85%**
Clinical detection, per year	2%	**2%**

Reference screening: Indonesia / Uganda. Scale-up ramps linearly from 2026 to 2050. Results are reported for 2026–2050.

Prophylaxis is assumed to slow progression by 55% through age 39, with no direct effect on mortality

Assumed in the model

On BPG, A$\rightarrow$B and B$\rightarrow$C hazards fall by 55% $\times$ adherence, at ages ≤ 39

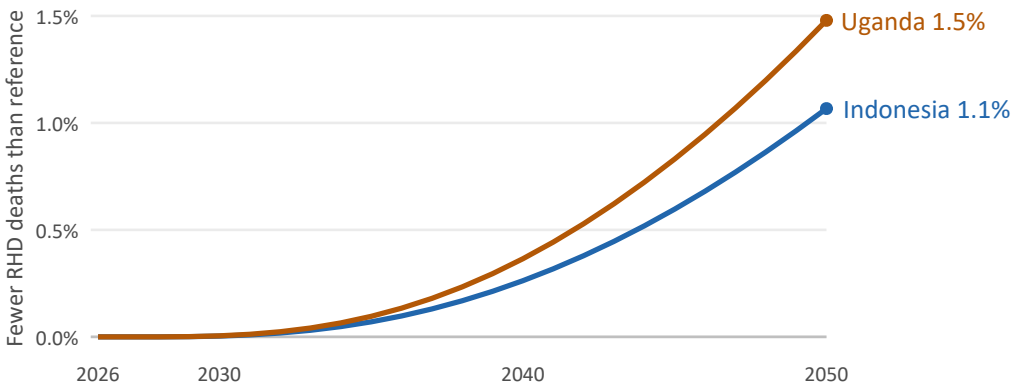
Adherence weight 0.70 $\rightarrow$ 0.90

Evidence it draws on

Recurrent rheumatic fever: RR 0.45 (0.22–0.92), one of three trials (Manyemba & Mayosi, 2002)

Latent RHD, ages 5–17: two-year progression 0.8% vs 8.2% (Beaton et al., 2022)

Scale-up averts about 1,500 RHD deaths in Indonesia and 150 in Uganda by 2050



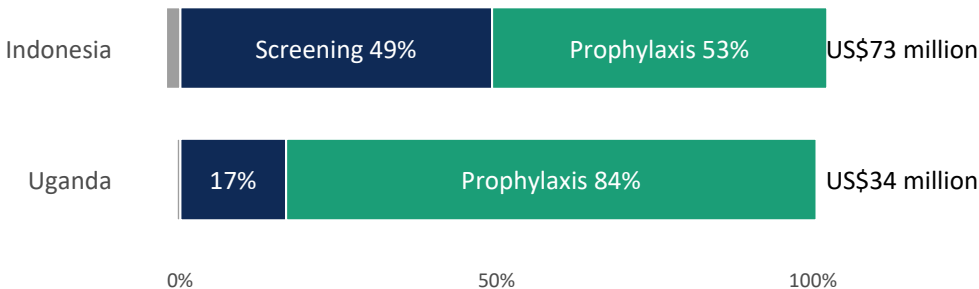
Annual reduction in RHD deaths, scale-up vs reference, 2026–2050. Cumulative: 0.3% (Indonesia) and 0.5% (Uganda) of reference-arm deaths; undiscounted.

In 2050, each extra screen-detected diagnosis takes about 590 screens in Indonesia and 170 in Uganda

Added by scale-up in 2050	Indonesia	Uganda
Echo screens	**4.70 m**	**1.78 m**
Screen-detected first diagnoses	**7,975**	**10,684**
First diagnoses, all routes	6,629	9,217
People on BPG (person-years)	52,000	93,044

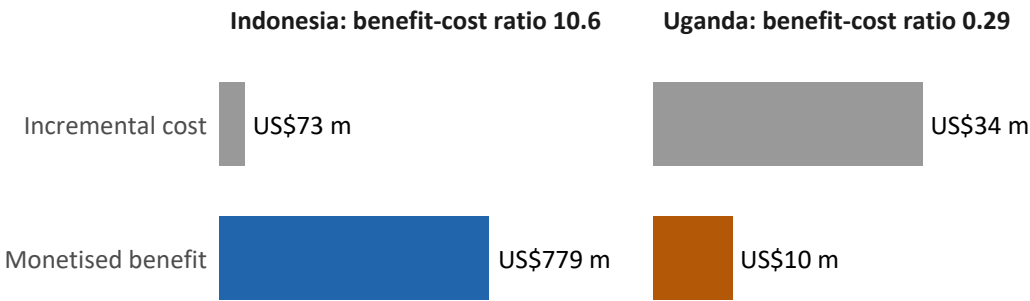
Scale-up minus reference, 2050 (final reporting year); m = million. Screens include people without RHD. Each screen-detected diagnosis needs one confirmatory echo.

Prophylaxis-years make up 53% of Indonesia's and 84% of Uganda's added cost through 2050



Incremental programme cost, scale-up minus reference, 2026–2050, discounted at 3% to 2026, constant 2025 US$. Dark: screening. Surgery: —2% and —1% (fewer stage C/D operations). Confirmatory echoes are not priced.

Through 2050, monetised benefits are 10.6 times costs in Indonesia but 0.29 times in Uganda



Discounted at 3% to 2026, 2026–2050. constant 2025 US$. Benefit = RHD deaths averted × value of a statistical life (VSL: reference-case default, PPP income elasticity 1.5, floor 20 × GNI per capita); m = million. Net benefit: US$705 million and —US$24 million.

Modelled deaths exceed GBD, so Indonesia's return is probably overstated

	Model / GBD deaths, 2019	Ratio as modelled	Ratio at GBD deaths
Indonesia	2.9	10.6	**3.6**
Uganda	1.7	0.29	**0.18**

All-age RHD deaths in the last calibration year. Last column: benefit–cost ratio divided by that excess, an illustrative rescaling that assumes the relative death reduction is unchanged; not a model run.

Returns hinge on progression speed, who gets BPG, and confirmation costs

- **Progression:** untreated adult B→C is 6.2% a year in Indonesia, 0.5% in Uganda
- **Targeting:** 49% (Indonesia) and 59% (Uganda) of 2050 BPG-years go to stage B over age 39, without modelled benefit
- **Confirmation:** about 222,000 added confirmatory echoes to 2050 are unpriced

Through 2050, screening plus prophylaxis is a slow investment: it repays its cost only where RHD progresses fast and kills often, as modelled for Indonesia but not Uganda.

Against GBD in 2019, the model under-counts prevalence and over-counts adult RHD deaths in both countries

Model / GBD target, 2019	Indonesia	Uganda
Prevalence, all ages	0.78	0.74
Prevalence, ages 0–19	1.06	0.95
Prevalence, ages 20+	0.75	0.67
RHD deaths, all ages	2.94	1.67
RHD deaths, ages 0–19	0.35	0.72
RHD deaths, ages 20+	3.39	1.81

Both sexes; GBD target = GBD rate × UN WPP population. Calibration fits incidence and six childhood transitions to GBD prevalence and deaths, 2000–2019 (scored from 2010).

Calibrated childhood progression is faster in Indonesia; adults progress at 0.48 times the childhood hazard

Annual probability, untreated	Indonesia	Uganda
A $\rightarrow$ B (childhood)	0.072	0.072
B $\rightarrow$ C, childhood	0.125	0.010
B $\rightarrow$ C, adult	0.062	0.005
C $\rightarrow$ D, childhood	0.250	0.171
C $\rightarrow$ D, adult	0.129	0.086
D $\rightarrow$ RHD death (fixed input)	0.060	0.060

Childhood = attained age ≤ 20; adult $= 1 - (1 - p_{\text{child}})^{0.48}$. Regression C$\rightarrow$B, B$\rightarrow$A and normalisation of A are also calibrated; D$\rightarrow$C is zero.

Both arms share population, clinical detection and surgery; only screening and BPG care differ

Reference → scale-up	Indonesia	Uganda
One screen, ages 5–39	5% → 80%	1% → 80%
BPG initiation, per year	50% → 90%	50% → 90%
BPG retention, per year	90% → 95%	90% → 95%
Good adherence share	60% → 85%	60% → 85%
Partial adherence share	20% → 10%	20% → 10%
Clinical detection, per year	2% → 2%	2% → 2%
Surgery coverage	5% → 5%	3% → 3%

Scale-up values reached in 2050 after a linear ramp from 2026. Screening sensitivity 90%; confirmation completion 90%; specificity taken as perfect. Partial adherence gives 50% of full protection.

Costs and benefits are discounted at 3% to 2026, and deaths averted are valued with a VSL that scales with income

Constant 2025 US$	Indonesia	Uganda
Echo screening encounter	US$1.10	US$0.52
Person-year on BPG	US$110	US$52
Valve surgery	US$8,998	US$6,560
Confirmatory echo	not costed	not costed
Price factor, assumed 2019 to 2025	1.000	1.312
GNI per capita, PPP, 2019	int$11,620	int$2,380
VSL / GNI per capita, 2019	70×	32×
Real income growth (carries VSL)	3.0%	3.2%
VSL in 2026	US$380,000	US$47,000

Unit costs: price year assumed 2019, converted with local CPI and exchange rates (World Bank WDI). VSL: U.S. US$9.4 million (2015) scaled by PPP GNI per capita, elasticity 1.5, floor 20 × GNI per capita.

Through 2050, scale-up costs US$47,700 per RHD death averted in Indonesia and US$225,400 in Uganda

2026–2050	Indonesia	Uganda
Incremental cost (discounted)	US$73.3 million	US$34.3 million
Monetised benefit (discounted)	US$778.7 million	US$10.1 million
Net benefit (discounted)	US$705.4 million	−US$24.2 million
Benefit–cost ratio	**10.6**	**0.29**
RHD deaths averted (undiscounted)	1,537	152
Cost per death averted	US$47,700	US$225,400
Cost per DALY averted	US$2,790	US$13,380

constant 2025 US$, summed to 2050 with the pipeline's definitions: cost per death = discounted cost / undiscounted deaths; DALYs = discounted deaths × 30 (years of life lost only). Deterministic; sensitivity analysis pending.

Annual incremental programme cost rises through the ramp, to US$10.5 million in Indonesia and US$5.8 million in Uganda

	Indonesia, 2025 US$			Uganda, 2025 US$		
Year	Ref.	Scale-up	Added	Ref.	Scale-up	Added
2030	41.6 m	42.8 m	1.2 m	7.3 m	7.7 m	0.4 m
2040	43.2 m	48.6 m	5.4 m	10.0 m	12.5 m	2.5 m
2050	41.9 m	52.4 m	10.5 m	12.6 m	18.4 m	5.8 m

Undiscounted annual programme cost (screening + BPG + surgery) by arm, constant 2025 US$; m = million. Confirmatory echoes are not priced.

Most RHD deaths averted by 2050 are among people aged 20-39

	Deaths averted		Share	
Age at death	Indonesia	Uganda	Indonesia	Uganda
0-19	92	16	6%	11%
20-39	839	79	55%	52%
40-59	606	57	39%	37%
60+	0	0	0%	0%

Cumulative RHD deaths averted, 2026–2050, undiscounted, by attained age at death. Protection on BPG applies at ages ≤ 39; deaths averted later reflect earlier slowing.

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